Final Project Report

UCI Adult (Census Income) — End-to-End ML Pipeline | Week 1, Day 5

1. Problem Definition

The goal of this project is to predict whether an individual's annual income exceeds $50,000 based on U.S. Census demographic and employment attributes (the UCI Adult / Census Income dataset). This is a binary classification problem relevant to applications such as targeted marketing, credit pre-screening, and socioeconomic research, where correctly identifying higher-income individuals — while controlling the cost of false positives and false negatives — matters more than raw accuracy alone.

2. Data Preparation

The dataset (48,842 rows) was loaded from OpenML (version=2) and split 80/20 into stratified train/test sets (39,073 / 9,769 rows, random_state=42) to preserve the class balance (23.93% positive rate) in both sets. Six engineered features were added based on Day 3's mutual information analysis: age and hours-per-week buckets, a capital-gain presence flag, a log-transformed capital gain (to handle its highly skewed distribution), a higher-education flag, and an education × hours interaction term. Preprocessing used median imputation and scaling for numeric features and most-frequent imputation with one-hot encoding for categorical features, applied via a ColumnTransformer fit only on the training data. Zero overlapping indices were confirmed between train and test sets, ruling out data leakage.

3. Model Development

Three model families were evaluated: Logistic Regression, Random Forest, and Gradient Boosting. Each was tuned independently before comparison.

Model	Best CV ROC AUC
Logistic Regression	0.9124
Random Forest	0.9163
Gradient Boosting (selected)	0.9277

Gradient Boosting outperformed both alternatives across all cross-validation metrics and was selected as the final model.

4. Model Tuning

Hyperparameters were searched using RandomizedSearchCV with 5-fold StratifiedKFold cross-validation for all three model families. The best Gradient Boosting configuration was: **learning_rate=0.05, n_estimators=200, max_depth=7, subsample=0.8**. This configuration balanced model capacity against overfitting risk, as confirmed by the diagnostics below.

5. Diagnostics

Learning curves were used to diagnose bias/variance trade-offs for both Logistic Regression (varying regularization strength C) and Gradient Boosting (varying max_depth), confirming the selected hyperparameters sit in a good bias-variance balance rather than under- or over-fitting. Probability calibration was assessed via calibration plots and Brier score; sigmoid calibration (CalibratedClassifierCV, 5-fold) was applied to align predicted probabilities with actual outcome frequencies. A threshold sweep (0.10 to 0.90) on calibrated probabilities identified **0.40** as the F1-maximizing decision threshold, which was used for all subsequent evaluation instead of the sklearn default of 0.5.

6. Final Results

The final model was re-evaluated on the untouched test set using predict_proba() with the selected 0.40 threshold applied manually (rather than the default-threshold .predict() method):

Metric	Value
Accuracy	0.8711
Precision	0.7299
Recall	0.7327
F1 Score	0.7313
ROC AUC	0.9292
PR AUC	0.8336
Brier Score	0.0868

Confusion Matrix (Test set, threshold=0.40):

	Predicted ≤50K	Predicted >50K
Actual ≤50K	6,797	634 (FP)
Actual >50K	625 (FN)	1,713

The Day 5 test ROC AUC (0.929) closely matches Day 4's CV ROC AUC (0.928), indicating strong generalization with no meaningful overfitting to the validation folds. The F1 score at the 0.40 threshold (0.731) is close to the ~0.781 F1 observed during the Day 4 threshold sweep, with the small gap expected since the sweep was measured on training data.

7. Model Interpretation

The strongest predictor by a wide margin is **marital-status: Married-civ-spouse** (importance 0.321), consistent with the well-documented correlation between marital status and household income in this dataset. **education-num** (0.102) and the engineered **edu_hours_interaction** feature (0.088) both rank highly, confirming the Day 3 feature engineering added genuine predictive signal. **log_capital_gain** and raw **capital-gain** (0.085 and 0.082) move in the expected direction — higher capital gains correlate with higher income probability.

Important caveat: fnlwgt (importance 0.042) is a Census Bureau sampling weight reflecting how many people in the population a row represents — it is **not a causal driver of income** and its importance score should be treated as a modeling artifact rather than a genuine insight.

8. Production Readiness

The final calibrated pipeline is saved as **final_pipeline.pkl** via joblib. A reusable **predict_income()** inference function loads the artifact once, accepts new raw data, applies all preprocessing and feature engineering automatically via the pipeline (no manual repetition), and applies the selected 0.40 threshold to return a final label. This was tested successfully on 10 held-out unseen examples. One deployment note: because the pipeline includes a FunctionTransformer wrapping the add_engineered_features() function, that function must be defined in the runtime namespace before the pickle is loaded.

9. Limitations & Future Improvements

- **fnlwgt** should ideally be excluded or down-weighted, since it is a non-causal sampling artifact rather than a genuine income signal.
- Error analysis shows elevated misclassification rates in **Protective-serv (21.5%)**, **Exec-managerial (19.2%)**, and **Craft-repair (18.1%)** occupations — likely because income in these roles varies with seniority/overtime not captured by current features. Adding occupation-specific interaction features could help.
- Sigmoid calibration slightly increased the Brier score on training data (0.0714 → 0.0868), reflecting the standard sharpness/calibration trade-off — worth revisiting with isotonic calibration or a larger calibration set in future iterations.
- The dataset reflects historical U.S. Census data and may not generalize to other populations, regions, or time periods without re-validation.
- Future work could explore fairness auditing across protected attributes, additional engineered interaction features, and ensembling Gradient Boosting with Logistic Regression for improved calibration.